Write UP

1. Bagaimana proses Exploratory Data Analysis (EDA) yang Anda lakukan? Apa saja insight menarik, pola data, atau kendala kualitas data yang berhasil diidentifikasi?

I conducted EDA in two stages. In the first stage, I audited the raw tables, checking for keys and joins, target distribution, time coverage, missing values, potential data leaks, and issues specific to each table. In the second stage, after the data was cleaned, I linked each table to the diagnosis.

Main results:

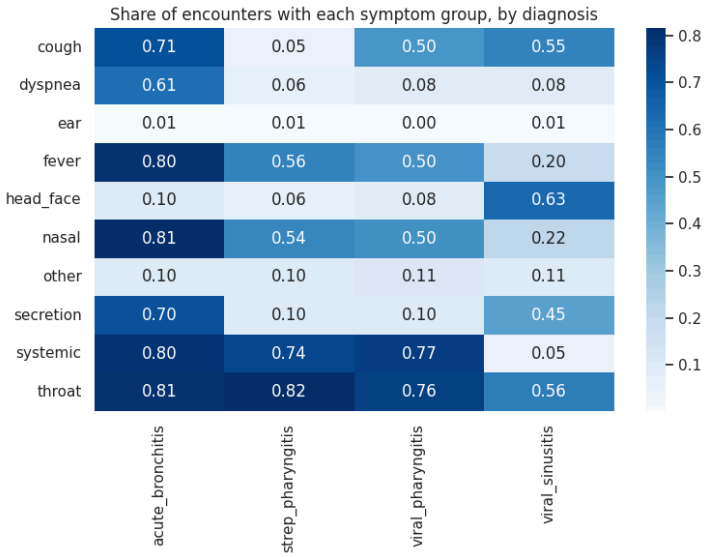
- Clean split: No patients appeared in both the training and test sets, so there was no leakage at the patient level.
- Target distribution: viral sinusitis 41%, viral pharyngitis 27%, acute bronchitis 26%, and strep only 5.4%.
- The data covers 2024–2026: The split is random, not temporal. No records fall after the prediction_time.

Table	Issue Found	Scale
patients	Age sentinel value 999	19 Patients
	Income in mixed formats: $, USD, thousand separators, k suffix	690 values
	Marital status mixes codes M and words Married	149 rows
	County typos	58 labels for 14 counties
	state is constant and city is 42% missing	whole columns
triage_vitals	Decimal commas	112 values
	Blood pressure in kPa and SpO2 as a fraction	618 BP rows and 431 SpO2 rows
	Values multiplied by 10 (heart rate 990)	103 values
	Values multiplied by 1.42 (SpO2 132.06 = 93×1.42	25 values

	BMI missing far more often than other vitals	37% vs 6%
triage_symptoms	Most raw labels are synonyms (118 raw to 70 after lowercasing and trimming)	689 rows with stray spaces
triage_sympyom_severity	765 Scores missing and dyspnea symptoms never recieve score	39 encounters with no severity at all
presenting_complaints	Only 8 fixed templates	5000 rows
history tables	Codes with several spellings of their name, missing names, fixed observation panels	up to 60 codes per table

Patterns related to the diagnosis:

Symptom groups separate three classes

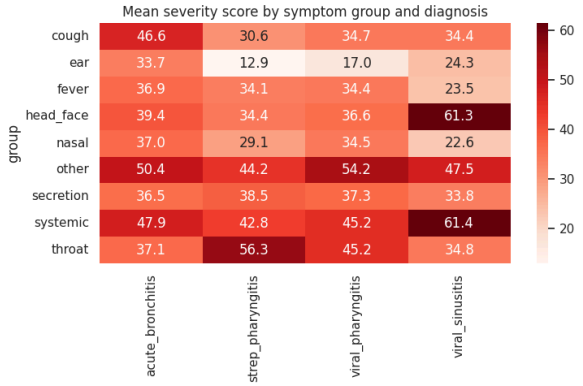


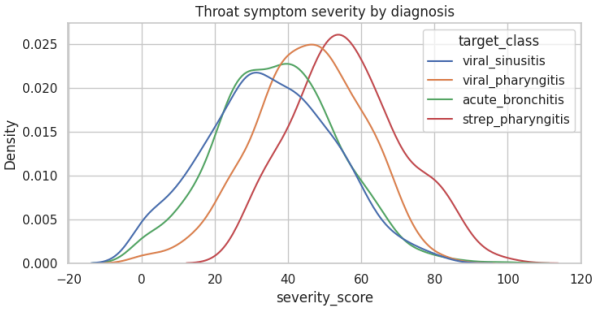
Groups of symptoms clearly distinguish the classes. Head and facial symptoms occur in 63% of cases of viral sinusitis, but only in 6–10% of cases in other classes. Systemic symptoms, on the other hand, are rare in sinusitis (5%) but common in other categories (74–80%). Acute bronchitis is often accompanied by dyspnea (61%) and secretions (70%). Strep patients almost never have a cough (5% vs. 50–71% in other categories).

Synonym choice is noise

The average number of distinct labels per group is nearly equal to the proportion of encounters that belong to that group.

Severity adds signal



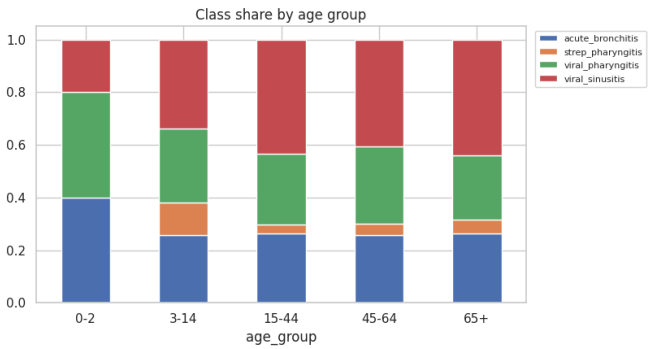


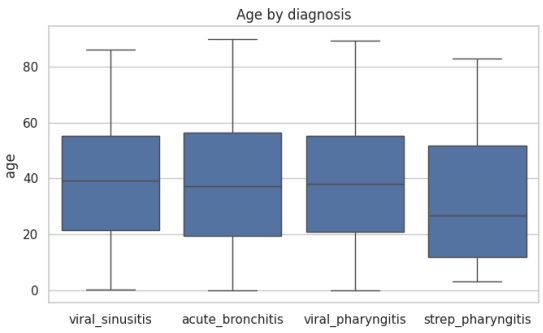
Severity provides additional information. Average throat severity: 56.3 for strep, 45.2 for viral pharyngitis, and around 35 for the other two categories. Sinusitis has the highest severity scores for the head, face, and systemic symptoms

Bronchitis patients report the most symptoms

Patients with bronchitis reported the most symptoms (a median of 6 different symptoms vs. 3 in other groups).

Age follows the Centor criteria





Strep infections most commonly affect children aged 3 to 14. This aligns with the Centor criteria, which assign +1 point for that age range.

Vitals barely differ between classes

Vital signs showed almost no difference between classes (the median differed by only 1–2 units). The proportion of kPa/fraction units was also the same across all classes (18.5–19.7%), so there was no label leakage.

History is weak

The history is relatively weak. Patients with bronchitis were slightly more likely to have a history of the condition.

Missing severity depends on the class

The absence of a severity score depends on the class. In bronchitis, 19.9% of symptoms had no severity score (vs. 8–11% in other classes), mostly because dyspnea was never assessed.

Presenting complaints repeat the symptom table

The complaints submitted almost always match the symptom table (the terms “cough,” “fever,” and “throat” match the symptom flags 100%).

2. Berdasarkan temuan EDA tersebut, bagaimana Anda merumuskan strategi preprocessing dan feature engineering? Lalu, bagaimana Anda memvalidasi bahwa tahapan ini efektif dalam meningkatkan kualitas data dan performa model?

Each preprocessing step is designed to address the findings from EDA. I validated this using before-and-after diagnostics and cumulative ablation across three seed folds.

EDA Findings	Actions	Inspection Results
Age 999	Ages > 110 treated as missing	79 missing ages
Mixed income formats	Parser removes $, USD, separators, and expands the suffix ‘k’	0 parsing failutes
Mixed marital status	map words to 1-letter codes	5 categories remaining
County typos	Lowercase + fuzzy match	58 to 14 labels with 0

	(cutoff 0.8) to 14 Massachusetts counties	mismatches
Decimal commas	Replace commas with periods	0 parsing failures
Mixed units	kPa×7.50062 to mmHg and SpO2 fraction × 100	Units consolidated
Values above the range	Try dividing by 10, then by 1.42. Save only if within range	103 rows corrected by dividing by 10 and 24 rows corrected by dividing by 1.42
Values below the range	Restore× 1.42 only if the result is an integer within the range	1 systolic value restored
Synonyms for symptoms	Lowercase + map 37 respiratory labels to 8 clinical groups	33 non-respiratory labels become ‘other’
Complaints repeating symptoms	Remove the complaints table	already proven to match 100% in EDA
History	Explicit cutoff filter + use codes (not names)	0 rows removed by filter

Tree-based models (LightGBM, XGBoost, CatBoost) accept missing values as-is because the missing patterns are informative. For logistic regression and random forests, I used median imputation plus a missing indicator in each CV fold, one-hot encoding for categorical variables, and standardization for logistic regression.

Features

Block	Count	Content
Symptoms and severity	23	sym_<group> flags, n_symptoms, n_resp_groups, sevmax_<group> for the 7 groups that have scores, overall mean and max severity, counts of symptoms without a severity row or score
Vitals	7	Six cleaned vitals and number of missing vitals
Demographics	8	Age, child_3_14, gender, income, race, ethnicity,

		marital status, and county
History	151	Record counts per table, days since the last previous visit, counts per visit type, and flags for codes recorded for at least 20 patients
Clinical	5	no_cough, centor_partial (no cough + fever + age 3 to 14), throat severity minus and divided by mean severity, and the group with the highest severity

I only retained the maximum severity per group (not the average) because a single encounter rarely has more than one label per group. Clinical blocks encode the Centor criteria that emerged in the EDA. Findings regarding tonsillar exudate and lymph node enlargement were not available in the data.

Leakage control: All features used only data prior to prediction_time. History codes were selected based on frequency (≥20 patients) without considering labels. The imputer and encoder were trained within CV folds.

Validation:

I tested LightGBM (fixed parameters, no tuning) on cumulatively added feature blocks. Score = out-of-fold macro F1, mean ± standard deviation across 3 seed folds.

Feature set	Features	Macro F1	Log loss
Symptoms and severity	23	0.683 ± 0.002	0.555
+ demographics	31	0.703 ± 0.005	0.534
+ vitals	38	0.704 ± 0.003	0.533
+ history	189	0.708 ± 0.003	0.496
+ clinical (final set)	194	0.710 ± 0.003	0.498
Final set + 87 raw symptom labels	281	0.710 ± 0.009	0.503

Summary of results:

- Symptoms alone accounted for 0.683. It is consistent with EDA findings.
- Demographics contributed the most (+0.020), primarily through age (children with strep).
- Vital signs contributed only +0.001 (still within the margin of error). They were retained because they are inexpensive and clinically relevant.
- Medical history adds 0.004 to F1 and reduces log loss quite significantly (0.533 to 0.496).
- Raw symptom labels do not improve the score. Instead, they increase variance. It justifies the grouping of synonyms.

3. Bisa ceritakan perjalanan eksperimen model Anda? Bagaimana proses validasi, tuning, hingga akhirnya memilih model final?

The final model is an average of a tuned LightGBM and CatBoost, with a score of 0.720 ± 0.003 below fully nested CV. I arrived at this result through a rigorous validation protocol, comparisons ranging from baselines to ensembles, and rules that reject gains smaller than the inter-seed noise.

Validation Protocol

- Stratified 5-fold cross-validation was repeated with 3 seeds (42, 7, 2024). I report the mean ± standard deviation across seeds. Differences below ~0.005 are within the seed-to-seed noise.
- All models used the same folds so that the OOF probabilities could be averaged without leakage.
- Imbalance was addressed in two ways: class_weight="balanced" during training, and post-hoc weighting on the strep probability before argmax. For the final candidates, I used full nesting (inner 4-fold within each outer fold).
- Primary metric: weighted macro F1. I also monitored raw argmax F1, strep F1, and log loss.

Comparison of models

All models have the same 194 features and the same folding mechanism (except for the compact model).

Model	Macro F1	Strep F1	Log loss
Dummy (class prior)	0.146	0.000	1.229
Logistic regression, all features	0.699 ± 0.004	0.372	0.522
Random forest	0.709 ± 0.005	0.404	0.537
LightGBM, baseline parameters	0.710 ± 0.003	0.372	0.498
LightGBM, tuned, class_weight balanced	0.710 ± 0.004	0.410	0.481
XGBoost	0.712 ± 0.005	0.388	0.442
LightGBM, tuned	0.713 ± 0.001	0.366	0.438
CatBoost	0.719 ± 0.002	0.396	0.448
Logistic regression, compact (29 features)	0.720 ± 0.005	0.410	0.464
Ensemble: LightGBM tuned + CatBoost (final)	**0.726 ± 0.004**	**0.408**	**0.439**
Ensemble + compact logistic regression	0.725 ± 0.008	0.402	0.437
Ensemble + both LightGBM variants	0.722 ± 0.001	0.394	0.443
Ensemble + strep specialist (α = 0.5)	0.729 ± 0.003	0.417	0.439

LightGBM tuned + per-group severity-missing flags	0.717 ± 0.006	0.388	0.438

The compact logistic regression model (only symptoms + age + clinical features) performed nearly as well as the tree model. This indicates that the signal is relatively simple and nearly linear.

Tuning

LightGBM was tuned using Optuna (60 trials, TPE sampler). The objective is log loss, not F1, because F1 is highly sensitive to even slight changes in strep predictions. Best results: shallow tree (5 leaves), learning rate 0.011, 937 trees. Log loss decreased from 0.498 to 0.438.

CatBoost uses depth 4, learning rate 0.03, L2 = 5, following the pattern observed in LightGBM. The number of iterations (1,644) was determined by early stopping.

Handling the imbalance

Handling class imbalance with class_weight="balanced" increased the raw F1 score but worsened the log loss. Post-hoc weights perform better because they preserve probability calibration (important when averaging). Under full nesting, the selected weights range from 2.1 to 3.8 across folds. For the test predictions, I used a weight of 2.4.

Rejected Experiments

- Strep specialist: Its AUC is lower than the ensemble's, and its best gain (+0.003) is still within the margin of error.
- Compact logistic regression as the third member: slightly lowered the mean and increased the variance.
- Flag severity-missing per group: only +0.004 with higher standard deviation.
- Raw symptom labels: did not improve the score while variance increased 3×.

Fully Nested Evaluation

Candidate	Screening macro F1	Nested macro F1	Nested strep F1	Difference
Ensemble: LightGBM tuned + CatBoost	0.726 ± 0.005	**0.720 ± 0.003**	0.406	−0.006
LightGBM, tuned	0.713 ± 0.001	0.718 ± 0.000	0.403	+0.005

The filtering protocol adds approximately ±0.006 in noise. I report the nested score as the final result.

FInal Choice

The ensemble (tuned LightGBM + CatBoost) has the highest nested score (0.720) and a log loss nearly identical to that of LightGBM alone. The advantage is very slight (0.002), so I consider both to be closely matched alternatives. LightGBM tuned alone is the second submission. For test predictions, each model was retrained on the entire training dataset with 5 seeds and then averaged. The public leaderboard scores (0.730 for the ensemble, 0.731 for LightGBM) align with the nested estimates.

4. Bagaimana hasil evaluasi akhir dari model tersebut? Di area mana saja model masih kurang akurat, dan apa faktor penyebab serta keterbatasannya?

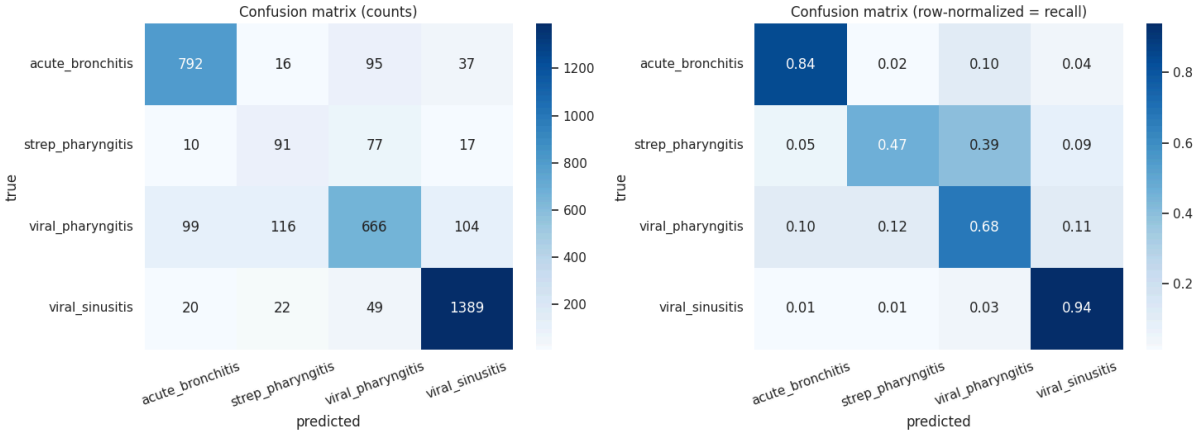
On the fold of seed 42 (with full-nesting weighting), the final model achieved a macro F1 score of 0.724 and an accuracy of 81.6%. Sinusitis and bronchitis were relatively reliable, while most of the remaining errors occurred between the two pharyngitis classes.

Per-class performance

Class	Precision	Recall	F1	ROC-AUC (one vs rest)	PR-AUC	Cases
viral_sinusitis	0.898	0.939	0.918	0.984	0.976	1,480

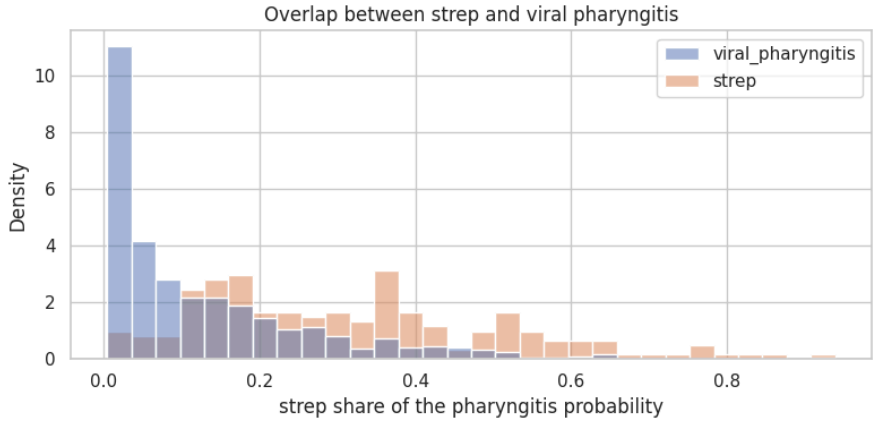
acute_bronchitis	0.860	0.843	0.851	0.968	0.930	940
viral_pharyngitis	0.751	0.676	0.712	0.922	0.796	985
strep_pharyngitis	0.371	0.467	0.414	0.920	0.409	195

Strep has an ROC-AUC of 0.920 (a good ranking), but its PR-AUC is only 0.409 due to its low prevalence (5.4%). From the confusion matrix, viral pharyngitis is most often misclassified as strep (116), sinusitis (104), and bronchitis (99).



Error 1: Strep vs. Viral Pharyngitis

The model misclassified 77 out of 195 strep cases (39%) as viral, and 116 viral cases as strep. The ROC-AUC for strep versus viral among pharyngitis patients is 0.812.



Group	Cases	No cough	Age 3–14	Mean age	Throat severity	Systemic symptoms	Centor (partial)

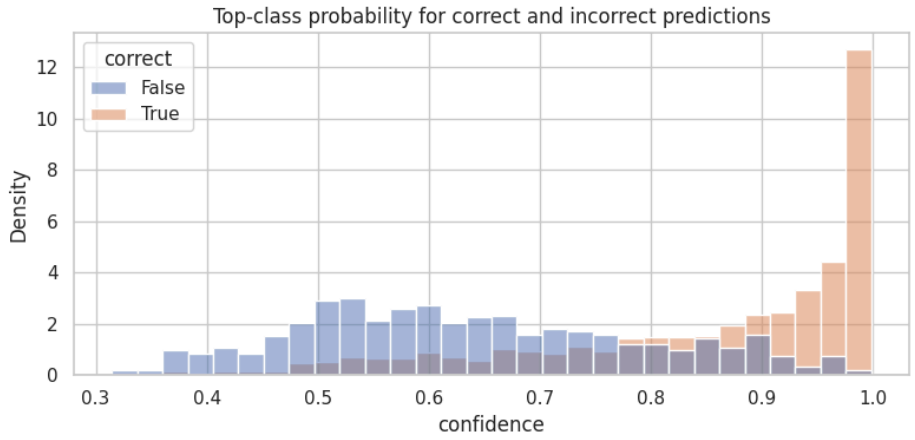
Strep, correct	91	100%	55%	28.8	63.9	71%	2.16
Strep missed as viral	77	95%	16%	40.8	49.5	94%	1.61
Viral, correct	666	48%	12%	39.5	44.5	87%	1.06
Viral predicted as strep	116	100%	44%	29.6	55.4	72%	2.10

The error patterns follow clinical logic: missed strep cases typically involve adults with systemic symptoms and moderate sore throat. Conversely, cases misdiagnosed as strep when they are actually viral typically involve children without a cough and with severe sore throat. The model applies logic similar to Centor's. Strep recall is high in children aged 3–14 (0.66), but low in adults (0.24–0.46).

Error 2: Bronchitis vs. Viral Pharyngitis

Of the 95 cases of bronchitis predicted to be viral, only 11% had dyspnea and 13% had secretions (vs. 70% and 78% in correctly identified bronchitis cases). Conversely, 99 viral cases predicted as bronchitis had dyspnea in 36% and secretions in 51%.

Belief as a Triage Signal



Top-class probability	Accuracy	Encounters
above 0.9	97.4%	1,627 (45%)
0.7 to 0.9	82.3%	1,015

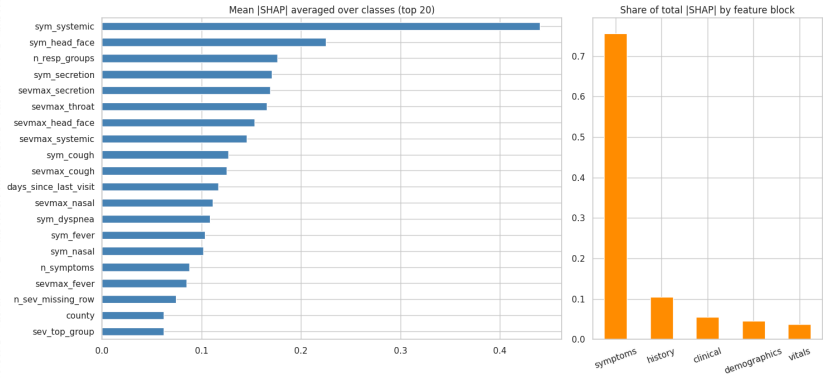
0.5 to 0.7	58.0%	750

Nearly half of the predictions are highly confident (97.4% accuracy). Clinics can accept confident predictions and refer uncertain cases to a doctor.

Interpretability (SHAP)

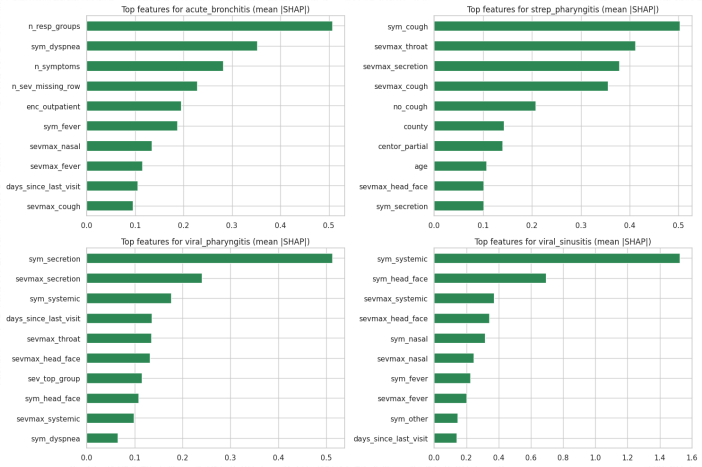
I retrained the tuned LightGBM on the entire training dataset and calculated the SHAP values using LightGBM's built-in TreeSHAP.

Global importance



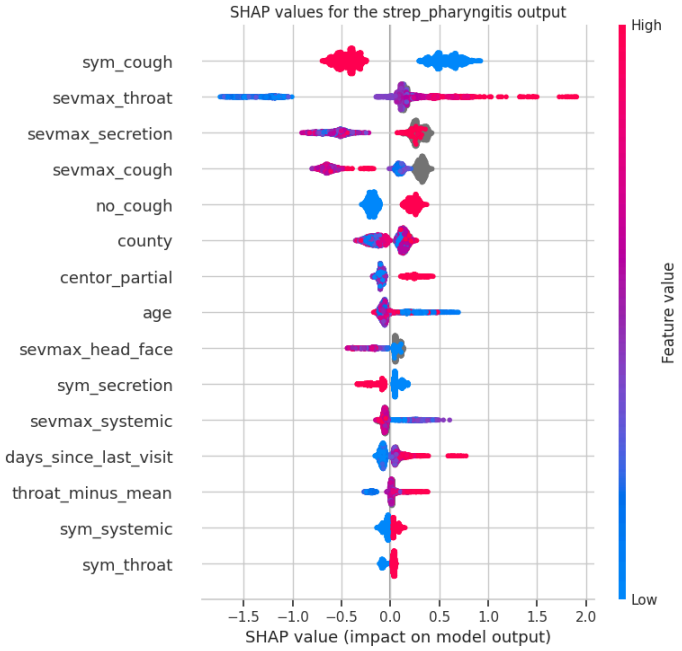
Symptom and severity features accounted for 75.6% of the total absolute SHAP values, medical history for 10.5%, clinical features for 5.6%, demographics for 4.6%, and vital signs for 3.7%. The presence of systemic symptoms ranked first, followed by head and face symptoms and the number of respiratory groups. Gain importance yielded the same top features.

By class (Figure 9)



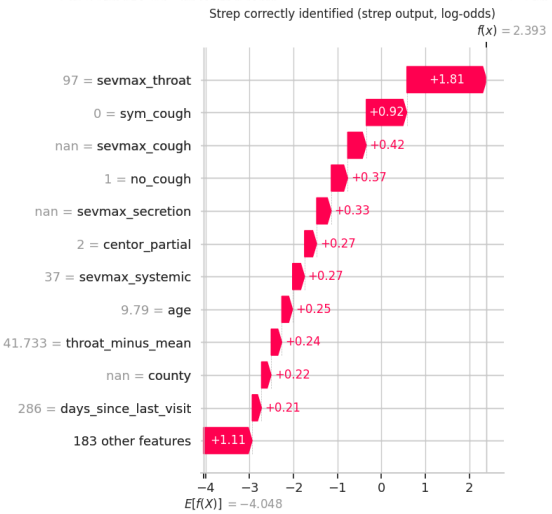
Sinusitis depends on the absence of systemic symptoms and the presence of head and facial symptoms. Bronchitis depends on the number of respiratory clusters, dyspnea, and the number of symptoms. Strep depends on cough, throat severity, secretion severity, no_cough, centor_partial, and age. Viral pharyngitis depends on the absence of secretions.

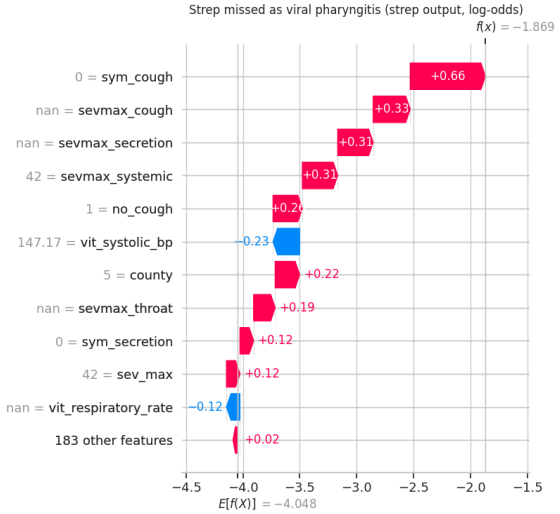
Direction of effects for strep



The absence of a cough adds 0.5 to 0.9 log-odds, severe throat symptoms add up to 1.9, and mild symptoms subtract up to 1.6. Younger age favors a strep diagnosis.

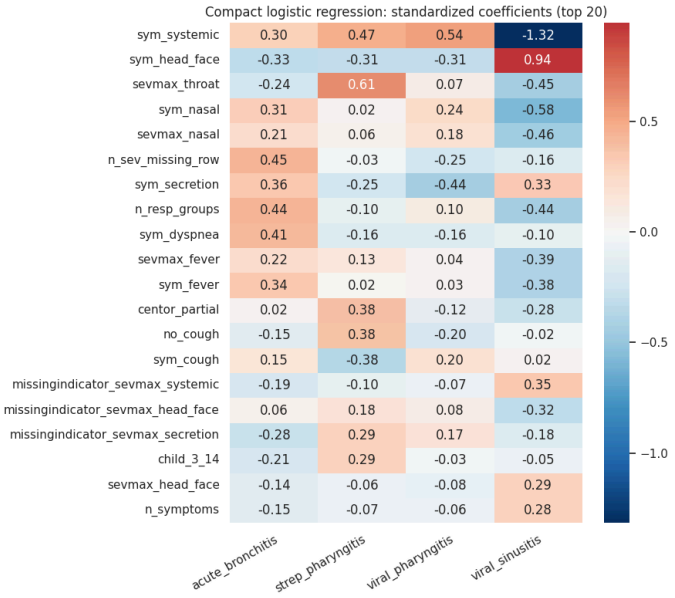
Two patients





For a 9.8-year-old child correctly identified with a throat severity score of 97 and no cough, throat severity alone adds 1.81 log-odds, and the output rises from a baseline of −4.05 to +2.39. For the strep case misclassified as viral, the absence of a cough helped (+0.66), but throat symptoms had no severity score. The model never detected the strongest strep signal for this patient, and the output remained at −1.87.

Transparent cross-check



The standardized coefficients from the compact logistic regression tell the same: systemic symptoms at −1.32 and head and face symptoms at +0.94 for sinusitis, throat severity at +0.61, no_cough and centor_partial at +0.38 each, and child_3_14 at +0.29 for strep. Two model families that agree on clinically plausible drivers make the model more credible.

Limitations and Reliability

- False positives are possible. The county ranked sixth for strep cases in SHAP, and the long interval since the last visit may have contributed to some of the strep cases. Neither of these factors has a clinical explanation.
- Residual corruption. There is still a possibility that vital sign values within a range multiplied by 1.42 have not been detected.
- Small minority classes & noisy estimates. The F1 score for strep varies by several points across seeds. The weighted screening protocol deviates by ±0.006 from full nesting. The public leaderboard has only about 22 strep cases. Therefore, decisions

are made based on nested CV, nested scores are reported as the final results, and differences below ~0.01 are treated as ties.

5. Jika memiliki waktu pengerjaan yang lebih luang, inovasi atau pendekatan baru apa yang ingin Anda coba untuk mengoptimalkan hasil ini? Apa alasannya?

Further Tuning Will Not Significantly Change This Model. What has a greater impact is new information about patients with pharyngitis and decision rules designed for clinical practice. Three findings support this conclusion: the 29-feature logistic regression model nearly matched the ensemble, all experiments following CatBoost remained within the noise seed, and both the ensemble and the strep-specific model plateaued at an AUC for strep versus viral of approximately 0.81. Further recommendations:

- Collect missing Centor/McIsaac items: Tonsillar swelling/exudate, tender anterior cervical lymph nodes, and measured body temperature are signs that distinguish strep from viral pharyngitis. Rapid antigen test results (if available) will be the strongest single input. Error analysis shows that missed cases are precisely those that could have been resolved by these signs.
- Replace the optimal F1 threshold with a clinical threshold: Missed strep cases carry a risk of complications (e.g., rheumatic fever), while false positives only require a rapid test. Set a threshold to achieve a specific recall target (e.g., 80%), then refer flagged patients for testing and report how many tests were required.
- Learn directly from relational structures: If patient histories are more comprehensive, graph or relational deep learning models based on patient–visit–code relationships could replace manually created count features and flags.

Note: Boleh menyertakan gambar